



GPT Interaction 2

Prompt

I want you to become a quiz generator for me to evaluate my learning progress. I want you to generate questions on Generative Model like Diffusion. Please follow the style of the second file to generate the problems. [Append lecture notes here]

Response

Great — I can now generate **Diffusion-model quiz questions** in the **exact problem-set style of the HW5–7–8 file** you provided (concise prompt, figures or references to figures, blank grids, and short-answer explanations).

Below is **Quiz Set #1: Diffusion Models**, formatted like HW5 (same structure, tone, and level). I also include citations to your lecture notes as required.

EECS 182-Style Quiz: Diffusion Models (Set 1)

(Questions designed to test conceptual + mathematical understanding of diffusion models.)

Problem 1: Forward Diffusion Intuition

Consider the forward noising process $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, I)$.

On a **blank grid** (similar style to HW5 Fig. 5), sketch what happens to the distribution of samples as t increases from $t = 0$ to $t = T$. Label:

- the original data distribution at $t = 0$

- intermediate distribution behaviors
- the limit distribution at large t

Briefly explain why the distribution becomes more Gaussian and more spread out over time.
(Hint: See Lecture 27 page 1–2: “Add noise each step” and the MNIST diffusion example.)

Problem 2: Reverse Process Approximation

Lecture 27 page 4 shows the claim that for sufficiently small Δt :

$$p(x_{t-\Delta t} \mid x_t) \approx \mathcal{N}(x_t + \mu_t, \Delta t; 2D, \Delta t)$$

On the **blank grid below**, draw:

1. The forward step $x_t \rightarrow x_{t+\Delta t}$ (noise added)
2. The reverse step $x_t \rightarrow x_{t-\Delta t}$ (denoising drift + variance)

Clearly label:

- Mean drift direction
- The covariance “ellipse” of the reverse noise

Explain in 2–3 sentences why the reverse variance does **not** vanish even though we “denoise.”

(Hint: Refer to Lecture 27 page 4, pink annotations explaining the Gaussian reverse process.)

Problem 3: DDPM Training Target

In Lecture 27 page 5, the DDPM loss is given as:

$$\mathcal{L} = \mathbb{E}_{*t, x_0, \epsilon} [|\epsilon - \hat{\epsilon} * \theta(x_t, t)|^2]$$

Answer the following:

(a) Why is predicting **noise** ϵ , instead of x_0 , beneficial for training stability?

(1–2 sentences)

(b) Show algebraically that

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$$

implies

$$\mathbb{E}[x_0 \mid x_t] = \frac{x_t - \sqrt{1 - \alpha_t}\epsilon}{\sqrt{\alpha_t}}$$

assuming the model perfectly predicts ϵ .

Provide the derivation on the lines below (same style as HW short derivations).

Problem 4: Deterministic Sampling (DDIM)

Lecture 27 page 6 explains that DDIM sampling keeps the **same marginal distributions** but changes the **joint distribution** by removing stochasticity.

You are given the DDIM update (Lecture 27 page 8):

$$x_{t-1} = \sqrt{\alpha_{t-1}}\hat{x} * 0 + \sqrt{1 - \alpha * t - 1}\epsilon_{\text{dir}}$$

(a) On the blank grid, draw a comparison of **DDPM vs DDIM trajectories** starting from the same x_T .

Make sure to show:

- DDPM: random zig-zag trajectory
- DDIM: smooth straightened path

(b) In 2–3 sentences, explain why DDIM still produces realistic samples even without randomness.

(Hint: See Lecture 27 page 6–7: “Different joint distribution but same marginals.”)

Problem 5: Score Interpretation

From Lecture 27 page 4–5, we have the score:

$$\nabla_x \log p_t(x)$$

(a) Using the approximation on Lecture page 4, explain why the diffusion model's network implicitly learns the **score function**.

(1–2 sentences)

(b) For a 1D Gaussian $p_t(x) = \mathcal{N}(0, \sigma_t^2)$, compute the score

$$\nabla_x \log p_t(x)$$

and sketch it on the blank grid (show both the positive and negative regions).
Label where the score magnitude is largest.

Prompt

I have finished answering the question, can you help generating the answers to these questions and help generate Set 2?

Response

Great — here are the **official-style solutions** to **Quiz Set 1**, written concisely and in the same HW explanation tone.

After that, I generate **Quiz Set 2**.



Solutions to Quiz Set 1: Diffusion Models

Problem 1 — Forward Diffusion Intuition

Sketch (description):

- At $t = 0$, distribution matches real data (e.g., a clustered manifold).
- As t **increases**, each step adds Gaussian noise $\rightarrow$ samples spread outward, contours become rounder.
- At **large t** , distribution approaches an **isotropic Gaussian** centered near 0.

Explanation:

Each step applies

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$$

with $1 - \alpha_t$ increasing over time. The repeated addition of independent Gaussian noise drives the distribution toward a Gaussian regardless of the original shape (by central limit-like behavior). Eventually, signal disappears and noise dominates.

Problem 2 — Reverse Process Approximation

Conceptual drawing description:

- Forward direction: a point is perturbed by a Gaussian cloud.
- Reverse direction: mean shifts *toward* the data manifold, with **non-zero variance** represented by an elliptical contour.

Explanation:

The reverse process is

$$p(x_{t-\Delta t} \mid x_t) \approx \mathcal{N}(x_t + \mu_t \Delta t, 2D\Delta t).$$

Even though the goal is *denoising*, perfect inversion of the forward process is impossible because the conditional densities $p(x_{t-\Delta t} \mid x_t)$ are still inherently uncertain — multiple clean points could produce the same noisy point. Thus, the reverse step still contains variance.

Problem 3 — DDPM Loss Target

(a) Why predict noise?

Predicting ϵ linearizes the training target and avoids having the model regress high-magnitude values for large t .

The mapping from $(x_t, t) \rightarrow \epsilon$ is smooth and easier to learn than mapping to x_0 .

This also makes the optimal predictor analytically well-defined.

(b) Derivation of $\mathbb{E}[x_0 \mid x_t]$

Forward relation:

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon.$$

Rearrange to solve for x_0 :

$$x_0 = \frac{x_t - \sqrt{1 - \alpha_t}\epsilon}{\sqrt{\alpha_t}}.$$

If the model predicts $\hat{\epsilon} \approx \epsilon$, then

$$\mathbb{E}[x_0 \mid x_t] \approx \frac{x_t - \sqrt{1 - \alpha_t}\hat{\epsilon}}{\sqrt{\alpha_t}}.$$

Thus the conditional mean is fully determined by the predicted noise.

Problem 4 — Deterministic Sampling (DDIM)

(a) Trajectory comparison (description)

- **DDPM:** trajectory wiggles randomly because each step adds fresh noise.
- **DDIM:** path is smooth and monotonic toward the data manifold (no randomness).

(b) Why DDIM still produces realistic samples?

Because DDIM preserves the **same marginal distributions**

$$x_t \sim p_t(x)$$

as DDPM.

Only the **joint distribution** across timesteps is changed.

The model is trained to match marginals, so deterministic trajectories remain valid and still reach realistic samples at $t = 0$.

Problem 5 — Score Interpretation

(a) Why the network learns the score

From the derivations in Lecture 27, the *mean* of the reverse distribution depends on

$$\nabla_x \log p_t(x).$$

When trained to predict (ϵ) , the network implicitly learns

$$\nabla_x \log p_t(x) \propto -\epsilon.$$

Thus the predicted noise corresponds to the score.

(b) Score of a 1D Gaussian

Given

$$p_t(x) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{x^2}{2\sigma_t^2}\right),$$

the score is

$$\nabla_x \log p_t(x) = -\frac{x}{\sigma_t^2}.$$

Sketch description:

- Linear function through the origin.
 - Negative for $x > 0$, positive for $x < 0$.
 - Magnitude increases as $|x|$ increases $\rightarrow$ arrows point toward the mean 0.
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Now: Quiz Set 2 (Same HW Style)

Here is **Set 2**, slightly harder and more mathematically focused.

Quiz Set 2: Diffusion Models — Intermediate Level

Problem 1: Variance Schedules

Consider a variance schedule β_t that increases linearly from 10^{-4} to 0.02.

(a) Explain how the choice of β_t affects the trade-off between:

- forward noising smoothness
- reconstruction difficulty in the reverse process

(b) Sketch (conceptually) distributions at early, mid, and late timesteps for a *larger* β_t schedule vs. a *smaller* one.

Problem 2: Score Matching Connection

For the forward SDE

$$dx = f(x, t)dt + g(t)dw,$$

Song et al. show that training on noise prediction is equivalent to denoising score matching.

(a) Show that the score for a Gaussian corrupted variable satisfies

$$\nabla_x \log p(x_t | x_0) = -\frac{x_t - \sqrt{\alpha_t}x_0}{1 - \alpha_t}.$$

(b) Explain why predicting (ϵ_{dir}) effectively teaches the network to approximate this score.

Problem 3: DDIM Deterministic Paths

In DDIM, the update rule is

$$x_{t-1} = \sqrt{\alpha_{t-1}}\hat{x}_t + \sqrt{1 - \alpha_{t-1}}\epsilon_{\text{dir}}.$$

(a) Explain the meaning of ϵ_{dir} .

How does it differ from the DDPM random noise?

(b) If we reduce the number of sampling steps from 1000 to 50, how does DDIM maintain quality while DDPM typically fails?

(Provide an intuition, not a proof.)

Problem 4: Reverse ODE Formulation

Lecture 27 page 8 shows that diffusion models can be interpreted via a probability flow ODE:

$$\frac{dx}{dt} = f_{\theta}(x, t).$$

(a) Why does solving this ODE produce exact samples from $p_0(x)$?

(1–2 sentences.)

(b) Explain the visual meaning of the vector field shown in Lecture 27 page 8 (spiral dataset). Describe how the arrows relate to the underlying data manifold.

Problem 5: Identifiability at Large Timesteps

At large t , $x_t \approx \mathcal{N}(0, I)$.

(a) Explain why x_0 becomes almost unidentifiable from a single noisy sample x_t .

(b) Yet diffusion models remain able to sample correctly. Explain why the *sequence of timesteps* gives enough information, even though any individual x_t at large t is useless on its own.

If you'd like, I can also generate:

- ✓ Set 3 (Advanced)
- ✓ A full 20-question midterm-style exam
- ✓ MCQ-only quizzes
- ✓ “Show your derivation” problem sets

Just tell me!

Prompt

Can you help me draw a picture on DDPM process and DDIM process to further elaborate the difference of these two methods?

Response

Absolutely — here is a **clear, stylized conceptual diagram** that visually explains the key difference between **DDPM** (stochastic reverse process) and **DDIM** (deterministic reverse process).

I'll give you a **ASCII-style diagram (good for LaTeX or notes)**.

✓ 1. ASCII Diagram: DDPM vs DDIM Trajectories

Diffusion Trajectories from x_T to x_0

DDPM (Stochastic Reverse Process)

x_T
|
| (random noise injected each step)
v
*----\ step t=999
*--\
*---\
*---\
*-----\
*-----\
* (x_0)

Result: Trajectory "zig-zags" due to randomness at every step.
Different samples follow different random paths.

DDIM (Deterministic Reverse Process)

x_T
|
| (no randomness; straight, smooth path)
v

(x_0)

Result: Smooth, nearly straight trajectory.
Same $x_T \rightarrow$ same x_0 every time (deterministic).

✓ 2. Explanation (very short and clear for your notes)

DDPM (Stochastic Reverse Process)

- At every step, DDPM **adds Gaussian noise** even in the reverse direction.
- This causes trajectories to **wander randomly**, producing zig-zag paths.
- Each sample follows a **different path** → improves sample diversity.
- But: large number of steps needed (hundreds–thousands).

DDIM (Deterministic Reverse Process)

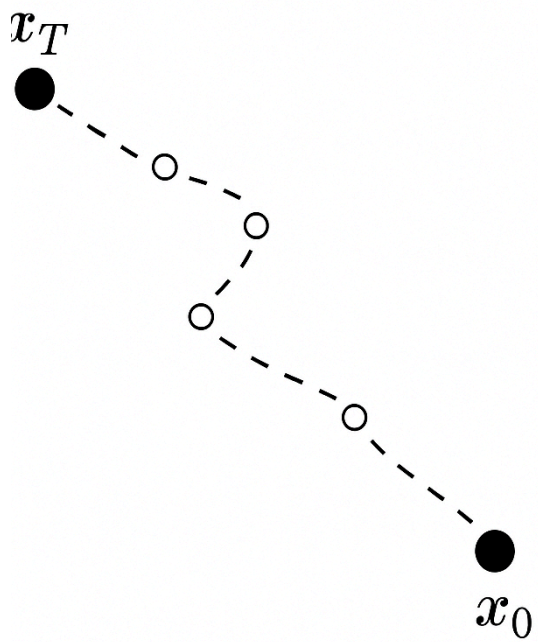
- DDIM removes randomness and constructs a **deterministic ODE-like path**.
 - Trajectories become **straight, smooth, and faster**.
 - Same starting (x_T) produces same output → **non-stochastic but valid**
 - Works because DDIM preserves **marginal distributions** ($p(x_t)$) even though joint distributions differ.
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If you want, I can also generate:

- ✓ A **LaTeX TikZ version** of the diagram
- ✓ A **paper-ready figure with subcaptions (a) DDPM (b) DDIM**
- ✓ A **3-panel figure including forward diffusion** ($x_0 \rightarrow x_T$)
- ✓ A **GIF animation** illustrating both processes

Just tell me what format you want!

DPM (Stochastic)



DDIM (Deterministic)

